

Who to Trust?

Phil 101 · Day 8

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Where we are

Last time

Two views:

1. Testimony is default justified, but this default can be overridden if you have positive evidence of dishonesty or unreliability.
2. You need positive reason to accept someone's testimony, but this reason can be really lightweight; it might just be that they passed a simple filter.

In practice, these will often coincide. Perhaps not always, especially in environments where you are really informationally constrained, but often.

The trouble with that

Either way, we assume that you can tell whether there is (a) positive evidence of unreliability/dishonesty, or (b) reason to think the person passes a simple initial screen.

That's a **very plausible** assumption for people who you know well.

It's not exactly as plausible for people you don't know.

And much of what we know, or think we know, comes from testimony from people we don't know.

You cannot check the work

To assess what climate scientists are saying in any detail, you need years of training.

The same arguably goes for vaccine effectiveness, or food safety inspection, or medical diagnoses.

This isn't because you are lazy, or ignorant. It's because it's *really hard* to become an expert in these fields, and even judging them requires specialist skill.

Simple checks

Now it is worth noting that there are some things you can check yourself.

- ▶ There are twice as many record high temperatures as record low temperatures.
- ▶ There were **zero** deaths of women aged 20-24 from cervical cancer in England from 2021 to 2024.

Occasionally there are really simple data points that help us non-experts. But while these are striking, the experts often use much more complicated tests.

But you still have to decide

You have to decide whether to get the vaccine, whether to vote for policies to mitigate climate change.

The people whose word you need to rely on are strangers whose backgrounds you do not know and whose work you cannot evaluate.

If you can't judge the science, what *can* you judge?

Anderson's answer

Judge the scientists instead

Elizabeth Anderson, who teaches philosophy here at Michigan, suggests that this is asking the wrong question.

i Anderson, p. 145

Most laypersons cannot directly judge the merits of most scientific claims. Instead, they mostly judge what to believe by judging whom to believe.

Instead of judging the **science**, we should judge the **scientists**. If they seem trustworthy, we should trust them.

Does this help?

This only helps if we can judge the scientists, and Anderson doesn't want to just presuppose that. So she goes into some detail about how a normal person can reasonably do that.

Can someone with no more education than a high school diploma, and just access to information freely available on the web, reasonably judge the scientific experts?

To be clear, there's a contrasting question where it's clear the answer is no: Can someone with no more education than a high school diploma, and just access to information freely available on the web, reasonably judge the expert scientific view?

How does changing the question help?

Three questions

Anderson says that we're all able to ask, and make some progress towards answering, these three questions.

Expertise

Is this person **in a position to know**? Do they have access to the evidence and the skills to evaluate it?

Honesty

Are they disposed to **communicate honestly** what they believe, and not to mislead by reporting only some of it?

Epistemic responsibility

Are they **responsive** to evidence and arguments that others raise against them?

Degrees of expertise

Anderson's point about expertise is that it comes in degrees, and the degrees are public.

Her list runs from laypersons at the bottom, through people with a science degree in some *other* field, through PhDs outside the field, PhDs inside it, then:

- ▶ researchers actively publishing in the field
- ▶ researchers whose work others in the field build on
- ▶ leaders: elected to academies, winners of the major prizes

The deference you owe someone rises as you go down, and Anderson says it rises **steeply** at the last three.

Being an expert in *something* need not count

A physicist with strong views about immunology is somewhere near the middle of that ladder, not the top.

Anderson notes that for a technical subject like climate science, someone with only a bachelor's degree or a PhD in an unrelated field has a *low* level of expertise.

All of which you can check without understanding any immunology. Biographical and publication information is on the web.

Everyone has some skills

- ▶ Many people have **situated knowledge**;
- ▶ Ordinary people are just as well placed as experts to make **value judgments**.

i Anderson, p. 146

Scientists have no special expertise on these value judgments; in a democratic society ordinary people are entitled to criticize scientific theories for failing to incorporate certain values – or for incorporating bad values.

Dishonesty as defeater

Anderson's list of things that cast doubt on honesty:

- ▶ conflicts of interest, such as funding from someone with a stake in the answer
- ▶ a record of dishonesty: plagiarism, faked data, citing research that doesn't say what they claim
- ▶ misleading statements: cherry-picked data, misused statistics, quotations out of context
- ▶ persistently misrepresenting opponents, or falsely accusing them of dishonesty

She concedes some dishonesty is hard for a layperson to spot. But funding sources and a history of retractions are a search away.

Dogmatic and arrogant

We can look at the character of individual experts.

Dogmatic

Persisting in a claim while ignoring counterevidence and counterarguments from people with relevant expertise.

Arrogant

Advancing those claims as things others should believe **on your say-so**, while refusing accountability.

Neither character is trustworthy. The dogmatist gives you no reason to think their claims track any evidence. The arrogant person is claiming an authority they haven't earned. (Of course, the person who keeps making false predictions and needs to change tack might not be great either.)

Signs of evasion

What does irresponsibility look like from outside?

- ▶ refusing to share data, or to give methods in enough detail for anyone to replicate the work
- ▶ taking the argument to the press or to politicians before making it to experts
- ▶ repeating claims that have been publicly refuted, without answering the refutation

That last one is given a name.

Dialogic irrationality

i Anderson, p. 147

Dialogic irrationality: continuing to repeat claims after they have been publicly refuted by others, without responding to the refutations.

The whale

Anderson's example

Here is an exchange of the kind she has in mind. Abridged, from page 148.

- ▶ **Sceptic:** There are no examples of transitional fossils between one species and another.
- ▶ **Paleontologist:** Consider the whale. We have a line of fossils starting from the four-legged *Sinonyx*, through *Pakicetus*, *Ambulocetus*, *Rodhocetus*, *Basilosaurus* and *Dorudon*, before we get to modern toothed whales — a gradual loss of hind limbs, and the nostrils migrating to the top of the head to form a blowhole.
- ▶ **Scep****ic:** There are no examples of transitional fossils between one species and another.

What you just did

You have no idea whether *Rodhocetus* is a transitional fossil. You didn't need to in order to think that Sceptic is behaving poorly.

i Anderson, p. 148

His response does not even have the *form* of a rational response – a point that can be grasped without being able to directly assess the *content* of the response.

The contrast

Now suppose the sceptic had said this instead:

Ambulocetus and *Rodhocetus* cannot be transitional fossils to modern whales, because they are reptiles.

That is almost certainly false. But at least it **looks like** a viable response; it's got the right **shape**. So it's immune to this particular criticism.

That's to say that the criticism is not meant to catch everything; it's meant to be simple to apply in a way that judging which fossils are reptiles is not.

Limits of the test

It's tricky though. What's the line between simply repeating your view without answering hard questions, and dismissing cranks as a waste of time?

We can try applying the same tests as before: it's fine to simply reject views from people far enough down 'the ladder'.

But there's a risk of circularity here.

You think my favorite scientist is dogmatic, because they don't answer a particular critic. I say that critic is a crank who can be safely dismissed. You say that I only think that because I'm too deferential to authority figures.

If I reply that I'm deferential to people with good practices, but I only think the practices are good because the people who are excluded have no authority, it's possible my attitude is circular.

iClicker QUIZ

A researcher publishes a result. She won't release her data. She published it in a journal she edits. Her funding is from a company that sells the treatment.

Which of Anderson's three checks has she failed?

- A. Expertise
- B. Honesty
- C. Epistemic responsibility
- D. More than one of these

Consensus

When they agree

The criteria so far tell you whom to trust. Anderson's last question is what to do when you've found them.

Disagreement at the frontier is normal, and she thinks it's *good*. Science works better when the people doing it come at problems differently. Since Anderson's paper came out, there has been a lot of work by philosophers of science on the importance of disagreements within science.

i Anderson, p. 149

Before a consensus, the best course for laypersons is to suspend judgment.

And after

Once a consensus of trustworthy experts is consolidated, she says you should accept it. That's true even if there are a handful of dissenters, which there should be!, and even if a few people in the consensus turn out to have erred or cheated.

Three things count as evidence that a consensus exists: surveys and meta-analyses of the literature, surveys of the trustworthy experts, and consensus reports from bodies like the National Academy of Sciences.

Do this yourself

Are the people teaching you trained properly? Does it include what they are teaching on? Do they acknowledge when they are more or less expert on things? Do they answer questions or simply repeat things?

These all seem like good questions. Note that they point in different directions. You want someone to acknowledge their weaknesses, but ideally not to have too many weaknesses in the first place. All that's ok; the point is to have **good** teachers, not **perfect** teachers.

Next week

What happens if you think, or have reason to think, that the institutions the experts are embedded in might be breaking down.

In particular, what happens when the machinery of science is deliberately attacked, and how do you tell a real disagreement from a manufactured one.